

Regional Digital Development and the Psychometric Network of School–Family–Classroom Experiences: Evidence from a Large-Scale Adolescent Survey

Abstract

Despite rapid expansion of digital infrastructure in education, it remains unclear whether macro-level digital opportunity contexts shape how adolescents' school–family–class experiences cohere as an interconnected system, beyond differences in average scores. Integrating digital divide perspectives with a meso-system view of schooling, we conceptualize regional digital development (*digi_level*) as a macro-level proxy for stratified opportunity and examine whether networks of parenting practices, school attitudes, identity, school problems, and class cohesion (items *as12*–*as16*) differ across contexts—particularly between mature and potential digital regions, with school sector as a benchmark. We expect higher digital maturity to be associated with more integrated network structures (i.e., stronger connectivity and lower fragmentation). Using TIGPS nationally representative data, we estimate EBIC-regularized partial correlation networks (Spearman-based EBICglasso), report centrality, and assess stability via nonparametric bootstrapping. Network differences are tested using the Network Comparison Test (NCT), where variations in global strength and edge structure indicate differences in system integration versus fragmentation. We argue that network models capture conditional dependencies and system organization not identifiable in latent-variable or multilevel approaches. Inference is limited to associational comparisons in cross-sectional data; regional classifications may reflect broader socioeconomic conditions. To address design considerations, we complement unweighted network estimation with survey-weighted regional summaries (TOTWGT), distinguishing structural from population-level inference. A fully reproducible pipeline is provided. Findings inform how relational configurations of school experiences vary across digital contexts and demonstrate a transparent workflow linking large-scale educational data, network psychometrics, and reproducible computation.

Keywords: Psychometric network, Gaussian graphical model, Digital development context, Adolescent

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